

10th grade

Learning evidence 2.1 Decision analysis supplementary activity

Assessment

This supplementary activity is assessed using the criteria listed below. To pass this supplementary work, all problems must correctly activate the assessment criteria. Passing a criterion requires it to be correctly activated within the work.

C2: Interprets decision alternatives, events, consequences, and states of nature.

C3: Builds a payoff table from a description of the problem.

C4: Explains the maximax criterion for decision making without probabilities.

C5: Summarizes the maximin criterion for decision making without probabilities.

C1: Describes the way a problem can be formulated for optimal decision making. Evaluates expected value.

Problems

P1 Smoothie stand menu planning

A pop up smoothie stand must choose a menu plan for a weekend festival. Plan A is a premium toppings menu and Plan B is a basic menu. Customer turnout can be high or low with probabilities 0,55 and 0,45. Profit outcomes are given directly. Under high turnout Plan A earns 700 and Plan B earns 480, and under low turnout Plan A earns 250 and Plan B earns 200. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

P2 Fundraiser T shirt design selection

A school fundraiser must choose a T shirt design plan for a seasonal sale. Plan A is a premium design, Plan B is a standard design, and Plan C is a budget design. Demand for shirts can be strong, moderate, or weak with probabilities 0,35, 0,45, and 0,20. Expected unit sales depend on demand and plan. Under strong demand Plan A sells 420 units, Plan B sells 520 units, and Plan C sells 600 units. Under moderate demand Plan A sells 300 units, Plan B sells 380 units, and Plan C sells 450 units. Under weak demand Plan A sells 180 units, Plan B sells 240 units, and Plan C sells 300 units. Unit prices are 22, 18, and 15 for Plans A, B, and C. Use these parameters to calculate the payoff table. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

P3 Streaming bundle selection

A streaming media service must choose a subscription bundle for the next quarter. Plan A is a premium bundle, Plan B is a family bundle, and Plan C is a value bundle. Demand can be surge, high, moderate, or low with probabilities 0,20, 0,35, 0,30, and 0,15. Expected subscriptions depend on demand and plan.

Under surge demand Plan A sells 500 subscriptions, Plan B sells 620 subscriptions, and Plan C sells 700 subscriptions. Under high demand Plan A sells 420 subscriptions, Plan B sells 520 subscriptions, and Plan C sells 600 subscriptions. Under moderate demand Plan A sells 300 subscriptions, Plan B sells 380 subscriptions, and Plan C sells 450 subscriptions. Under low demand Plan A sells 200 subscriptions, Plan B sells 260 subscriptions, and Plan C sells 320 subscriptions. Unit prices are 24, 19, and 16 for Plans A, B, and C. Variable costs per subscription are 11, 9, and 7. Use these parameters to calculate the profit payoff table. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

P4 Courier routing strategy choice

A bike courier cooperative must choose a routing approach for the next season. Strategy A uses an in house scheduling system and Strategy B uses a contracted routing partner. Delivery demand can be strong or weak with probabilities 0,55 and 0,45. Energy cost conditions can be favorable or unfavorable with probabilities 0,70 and 0,30, and these conditions are independent of demand. Revenue in thousands of dollars depends on demand and strategy. Under strong demand revenue is 620 for Strategy A and 660 for Strategy B. Under weak demand revenue is 340 for Strategy A and 320 for Strategy B. Operating costs in thousands of dollars depend on energy cost conditions and strategy. Under favorable costs operating costs are 260 for Strategy A and 300 for Strategy B. Under unfavorable costs operating costs are 340 for Strategy A and 380 for Strategy B. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

P5 Water utility treatment mix planning

A regional water utility must choose a treatment mix for the next year. Strategy A is filtration heavy, Strategy B is chemical treatment heavy, and Strategy C is balanced. Demand can be strong or weak with probabilities 0,55 and 0,45. Operating cost conditions can be favorable or unfavorable with probabilities 0,60 and 0,40, and these conditions are independent of demand. Revenue in thousands of dollars depends on demand and strategy. Under strong demand revenue is 680 for Strategy A, 620 for Strategy B, and 650 for Strategy C. Under weak demand revenue is 360 for Strategy A, 380 for Strategy B, and 370 for Strategy C. Operating costs in thousands of dollars depend on cost conditions and strategy. Under favorable costs operating costs are 300 for Strategy A, 280 for Strategy B, and 290 for Strategy C. Under unfavorable costs operating costs are 380 for Strategy A, 360 for Strategy B, and 370 for Strategy C. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?